Mustapha Ben-Cheikh

Data Science | Artificial Intelligence | Software Applications

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PROFILE

I build applied projects connecting data analysis and machine learning with web and mobile applications. My project work spans language models, computer vision, geospatial analysis, prediction, and optimization.

SELECTED PROJECTS

01 APPLIED AI | MOBILE

Smart Syndic AI

WhatsApp support assistant for property management, with a Flutter dashboard for residents, tickets, broadcasts, and analytics.

FastAPI | Twilio | ChromaDB | Flutter

02 NLP | RETRIEVAL

RAG Qwen

Retrieval-augmented question answering over course documents using ChromaDB and Qwen2.5-7B-Instruct.

Qwen2.5 | ChromaDB | Embeddings

03 DEEP LEARNING

Multimodal CTR Prediction

Combines CLIP item-image features and user behaviour sequences in a Deep Interest Network-style click predictor.

Python | CLIP | DIN

04 GEOSPATIAL

GeoAI Flood Risk

Streamlit prototype for building-level flood-risk analysis using building, river, road, and elevation layers.

GeoPandas | Streamlit | Folium

05 COMPUTER VISION

Satellite Image Inpainting

Context-aware GAN workflow for restoring masked satellite-image regions, evaluated with PSNR, SSIM, and LPIPS.

PyTorch | GAN | PSNR / SSIM

06 DATA | PREDICTION

Moroccan Air Quality

Air-quality exploration and prediction for Moroccan cities, including a simulated federated-learning workflow.

Python | scikit-learn | Jupyter

07 SECURITY | ML

DoS / DDoS Detection

Machine-learning classification of network-flow records with a Streamlit interface for CSV inference.

scikit-learn | XGBoost | Streamlit

08 OPTIMIZATION

PSO / MOPSO

Particle-swarm tuning for classifiers alongside multi-objective benchmarks and Pareto visualizations.

Python | scikit-learn | Plotly

MORE PROJECTS

COVID-19 Big Data Analytics | Mini COVID-19 Dashboard | Plagiarism Checker | Flask CSV Dashboard | Survey Data Analysis

TOOLS USED ACROSS THESE PROJECTS

Languages: Python, Dart | Machine learning: scikit-learn, XGBoost, PyTorch, CLIP
AI & retrieval: Qwen, RAG, ChromaDB, Sentence Transformers | Data: pandas, NumPy, Cassandra, PostgreSQL
Applications: Flask, FastAPI, Flutter, Streamlit | Geospatial: GeoPandas, Shapely, Folium, Plotly

Contact

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